

MUTUAL COUPLING REDUCTION IN MIMO PATCH ANTENNA ARRAY FOR Wi-Fi AND BLUETOOTH APPLICATIONS

A Project Report

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under the supervision of

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Certificate

This is to certify that this project report titled “**Mutual coupling reduction in MIMO Patch Antenna Array for Wi-Fi and Bluetooth applications**”, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, by **J Rohith, P Yashwanth, C Tej Siddanth**, is a bonafide record of work carried out by them, under my supervision and that it has not been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.

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We do hereby declare that this project report titled “**Mutual coupling reduction in MIMO Patch Antenna Array for Wi-Fi and Bluetooth applications**”, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, is a true record of work carried out by us, under the supervision of **Ms. Mekaladevi V** and that all information contained herein, which do not arise directly from our work, have been properly acknowledged and cited, using acceptable international standards. Further, we declare that the contents of this report have not been submitted, in part or in full, for the award of any other degree or diploma.

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Abstract

In this work, a compact 1×2 microstrip patch antenna array operating in the 2.4 GHz ISM band is designed for Wi-Fi and Bluetooth applications. To mitigate the mutual coupling between closely spaced antenna elements, a parasitic decoupling structure is introduced between the patches. The antenna design is developed and optimised using Ansys HFSS, where key parameters such as element spacing and the geometry of the decoupling structure are carefully tuned to improve impedance matching and isolation while maintaining stable radiation characteristics.

Simulation results show that the antenna resonates near 2.4 GHz with good return loss and achieves an isolation of approximately 17.4 dB between the elements. The array also maintains a broadside radiation pattern with a gain of around 7 dBi, making it suitable for short-range wireless communication systems.

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List of Abbreviations

ANSYS	Analysis System
CMA	Characteristic Mode Analysis
CP	Circular Polarization / Circularly Polarized
ECC	Envelope Correlation Coefficient
FR	Flame Retardant
FSS	Frequency Selective Surface
HEM	Hybrid Electric Magnetic Mode (dielectric patch mode)
HFSS	High Frequency Structure Simulator
IFA	Inverted F-Antenna
ISM	Industrial, Scientific, and Medical (band)
MCM	Mode Cancellation Method
MIMO	Multiple Input Multiple Output
PCB	Printed Circuit Board
PDSS	Parasitic Dollar-Shaped Structure
RT/Duroid	Rogers Teflon-based Duroid
SMA	Sub Miniature version A (connector type)
VNA	Vector Network Analyser
Wi-Fi	Wireless Fidelity

List of Symbols

f	Frequency
ε_r	Relative permittivity of the substrate
ε_{eff}	Effective dielectric constant
L	Patch length
W	Patch width
h	Substrate height
ΔL	Extension length due to fringing
L_{eff}	Effective length of the patch
c	Velocity of light
λ	Wavelength
f_r	Resonant frequency
ECC	Envelope Correlation Coefficient (MIMO performance metric)
NL	Neutralisation line

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with anti-parallel antenna
configuration

Fig.5.3.1.b

Simulated Gain plot with a 34
Parasitic element with anti-
parallel antenna
configuration

1. Introduction

Mutual coupling causes unwanted interaction between the antenna elements through electromagnetic fields. This leads to problems such as reduced isolation, poor impedance matching, and lower radiation efficiency. It can also increase the correlation between channels, which affects the diversity and capacity of the system. Therefore, minimising mutual coupling is a major challenge in designing efficient MIMO antenna arrays. In this project, a MIMO microstrip patch antenna array operating at 2.4 GHz is to be designed and analysed. The antenna is targeted for wireless communication applications such as Wi-Fi, Bluetooth devices, where compact size and high performance are essential. Simulation and optimisation are carried out using full-wave electromagnetic simulation software, followed by fabrication and testing to verify the results. The outcomes are expected to contribute toward the development of efficient antenna systems for wireless communication applications.

1.1 Motivation

In today's world, almost every modern device — from smartphones and smartwatches to routers and IoT modules — relies heavily on wireless connectivity. Multiple-Input Multiple-Output (MIMO) antenna systems have become a key solution to meet these needs by improving throughput and link quality without increasing bandwidth or transmit power. However, one of the major challenges faced by MIMO systems is mutual coupling between antenna elements, which can degrade overall system performance. This issue particularly becomes critical when antennas are placed close together in compact devices, as in most Wi-Fi and Bluetooth modules operating at 2.4 GHz.

The motivation for this project stems from the need to design antennas that can maintain high isolation while being compact and efficient. Since portable devices have limited space, keeping a sufficient distance between antenna elements is not always feasible. Therefore, developing effective decoupling techniques can significantly enhance the performance of MIMO systems. Understanding and reducing mutual coupling not only improves isolation and gain but also helps achieve a more stable radiation pattern and better signal integrity — all of which are crucial for practical wireless communication systems.

Ultimately, this project motivates us to contribute, even in a small way, to ongoing research in modern communication systems. By experimenting with decoupling structures and validating them through simulation and measurement, we aim to develop a simple yet effective design that can inspire future innovations in compact antenna systems. The satisfaction of designing, fabricating, and testing something that can actually improve wireless communication performance makes this project truly rewarding for us.

1.2 Challenges

One of the main challenges in designing a MIMO patch antenna array is achieving good isolation between antenna elements without increasing the overall size of the structure. Since most Wi-Fi and Bluetooth devices operate in compact enclosures, the antennas are usually kept very close to each other — often less than half the wavelength apart.

Another major difficulty lies in selecting and optimising the right decoupling structure. There are many techniques like slots, parasitic elements, and defected ground structures, each with its own pros and cons. Choosing the most effective one for 2.4 GHz operation involves understanding how the surface currents interact between elements and how each modification impacts bandwidth, isolation, and gain. Sometimes, even a small change in geometry or substrate parameters can shift the resonant frequency or degrade other characteristics, which makes the design process quite sensitive and time-consuming.

Simulating and validating the design using software like Ansys HFSS also poses challenges. Accurate modelling requires proper meshing, boundary setup, and excitation, which are easy to get wrong for beginners. Achieving convergence in results and interpreting the S-parameters correctly takes patience and practice. Additionally, once the simulation results are obtained, correlating them with real-world measurements can be difficult due to fabrication tolerances and losses in the substrate and connectors.

Finally, practical implementation adds its own hurdles. Fabricating the antenna with precise dimensions, ensuring good soldering for SMA connectors, and testing it in the anechoic chamber using a VNA demands both technical skill and attention to detail. Even minor errors during fabrication can cause deviations in performance compared to the simulated model. Overcoming these challenges is part of what makes the project both tough

and rewarding. It teaches us the importance of accuracy, iteration, and hands-on experimentation in real engineering work.

2. Literature Survey

In the paper Compact Self-Isolated Four-Element MIMO Antenna for WLAN and ISM Bands, operating in the 2.25–2.87 GHz range, which covers the WLAN and ISM 2.4 GHz bands. The antenna is based on printed dipole elements with integrated $\lambda/4$ balun feeding. What makes the design impressive is how it uses self-isolation through the smart arrangement of elements rather than relying solely on external decoupling networks. By placing dipoles orthogonally in a compact square arrangement, mutual coupling is naturally reduced, and then a simple printed decoupling element further suppresses residual coupling. For me as a student, this shows that effective antenna design is not always about adding complexity; sometimes, a thoughtful arrangement with the right feeding structure can achieve high isolation and efficiency while keeping the antenna simple and manufacturable.[1]

The work focus is microstrip patch antennas that were arranged closely with edge-to-edge spacing of just $0.016\lambda_0$ (≈ 2 mm at 2.45 GHz). The patches were probe-fed and operated in the 2.39–2.53 GHz band. Because of such close spacing, the antennas originally showed very poor isolation (about 5 dB). To solve this, a mode cancellation method (MCM) was introduced, where a lumped inductance or a small inductive strip was inserted between the patches. With this simple adjustment, the isolation improved dramatically to better than 15 dB, while maintaining stable radiation with a symmetric broadside beam and gain around 8.9 dBi. What stands out here is that a small feeding modification and a basic inductive element can fundamentally reshape how the fields interact. As a student, it taught me that sometimes the smallest, simplest additions, when guided by strong theory, can solve what looks like an impossible challenge in compact antenna design.[2]

The study examined a dual-band four-port rectangular patch at 2.35 GHz and 3.59 GHz. All four ports were excited in a co-polarised configuration using a standard probe-feed. The challenge with such designs is that using a single radiator with multiple ports usually results in very high mutual coupling. To overcome this, characteristic mode analysis (CMA) was applied, which guided the placement of shorting pins and ring slots on the patch. These structural changes altered the modal distribution of electric fields, effectively pushing field nulls closer to the unexcited ports. The result was a substantial reduction of mutual coupling to below -15 dB across both bands, while still maintaining the advantages of a low-profile,

single-layer, and single-element radiator. Importantly, the antenna preserved stable broadside beams and co-polarised radiation patterns, which are critical for practical communication systems. As a student, what I found inspiring is how advanced analysis tools like CMA can make antenna design more scientific rather than trial-and-error. By directly controlling how modes combine, engineers can achieve compact, dual-band, and multi-port antennas that are both efficient and reliable.[3]

The work focuses on reducing mutual coupling between two closely placed antennas in devices like laptops, where space is extremely limited. The proposed antenna system consists of two identical Inverted-F Antennas (IFAs) printed on an FR4 substrate of size $6.5 \times 41 \text{ mm}^2$, separated by just 2 mm. To minimize coupling, the authors introduced three innovative decoupling techniques: A decoupling inductor (L1) placed between the open ends of the IFAs to handle low-frequency decoupling (2.4 GHz band), A modified decoupling branch (L2) designed to improve isolation in the higher 5–6 GHz region, A modified neutralization line (L3) that ensures phase cancellation of mutual currents at higher frequencies. The optimised inductance values were $L1 = 3.6 \text{ nH}$, $L2 = 22 \text{ nH}$, and $L3 = 28 \text{ nH}$, tuned using HFSS simulations. These components helped the antenna achieve $|S_{21}| > 22 \text{ dB}$ isolation across Wi-Fi 6E frequency bands, which is a remarkable improvement compared to traditional dual-antenna systems. In terms of radiation, the antenna achieved peak gains of 4.27 dBi (2.4 GHz) and 4.43 dBi (5–7 GHz), with an efficiency above 40%. The Envelope Correlation Coefficient (ECC) remained below 0.3, confirming good MIMO performance. The radiation pattern was almost omnidirectional, suitable for laptop integration. Key Technical Details: Frequency Bands: 2.4–2.484 GHz, 5.15–5.825 GHz, 5.925–7.125 GHz, Antenna Type: Dual Inverted-F Antenna (IFA), Feeding Technique: Microstrip feed on FR4 substrate ($\epsilon_r = 4.4$, thickness = 1 mm), Isolation Achieved: $> 22 \text{ dB}$, Antenna Size: $6.5 \times 41 \text{ mm}^2$. Application: Wi-Fi 6E-enabled laptops and compact wireless systems.[4]

The major challenge addressed here is mutual coupling reduction between closely placed antenna elements. The proposed MIMO antenna uses two monopole radiators, each composed of a rectangular patch, an inverted-E-shaped arm, and an asymmetrical T-shaped strip. These elements work together to produce two resonant frequencies—2.5 GHz (for 5G Sub-6) and 4.5 GHz (for Wi-Fi 6E). The overall antenna size is just $45 \times 30 \text{ mm}^2$, with the two elements separated by 5 mm ($\approx 0.04\lambda$ at 2.5 GHz). The novelty of this design lies in a

Parasitic Dollar-Shaped Structure (PDSS) placed between the two monopoles. The PDSS is made up of two interlocking C-shaped loops and a vertical rectangular strip acting as a stop-band filter. This unique geometry suppresses surface waves, reducing mutual coupling to -15 dB. The antenna employs a Coplanar Waveguide (CPW) feed for better impedance matching and simple fabrication on an FR4 substrate ($\epsilon_r = 4.4$, thickness = 0.8 mm). Measurements confirmed good agreement with simulations, showing an efficiency of 70–75%, a gain of 2.6 dBi, and $ECC < 0.02$, which meets MIMO performance standards. Key Technical Details: Frequency Bands: 2.39–2.57 GHz (5G Sub-6), 3.82–6.95 GHz (Wi-Fi 6E), Antenna Type: Dual-band monopole MIMO antenna, Feeding Technique: CPW (Coplanar Waveguide) feed, Decoupling Structure: Parasitic Dollar-Shaped Structure (PDSS), Isolation Achieved: -15 dB across both bands, Antenna Size: 45×30 mm², Performance: Gain ≈ 2.6 dBi, Efficiency $\approx 70\%$, $ECC < 0.02$. [5]

The main motivation is to reduce mutual coupling in compact antennas, especially in in-band full-duplex (IBFD) communication systems where transmission and reception happen simultaneously at the same frequency. Traditionally, improving isolation between two ports of a microstrip antenna requires additional neutralisation lines. These methods increase complexity and size. However, this paper achieves excellent decoupling by cleverly combining two resonant modes of the same patch — mode 2 (TM₀₁) and mode 4 (TM₂₀) — to bend the electric-field nulls in such a way that each feed port lies near a null of the other, reducing the coupling naturally. To accomplish this, the authors use Characteristic Mode Analysis (CMA) to study and control how the modes interact. Further, small circular slots are cut near the feed ports to improve impedance matching. When tested, the proposed antenna operated efficiently between 5.30 and 5.54 GHz, maintaining $|S_{11}| < -10$ dB and an isolation of $|S_{12}| = -17$ dB, which is a significant improvement over the -3.5 dB of a conventional dual-port MPA. The antenna preserved a broadside radiation pattern with colinear polarisation, making it ideal for IBFD and MIMO systems that require high isolation and same-polarisation transmission. The measured gain was around 5.0 dBi with good agreement between simulation and experiment. Key Technical Details: Operating Frequency: 5.30–5.54 GHz, Antenna Type: Single-layer, dual-port Microstrip Patch Antenna (MPA), Feeding Technique: Probe-fed microstrip ports (two symmetrical feed points), Isolation Achieved: -17 dB, Peak Gain: ≈ 5.0 dBi, Structure Additions: Six shorting

pins and circular slots for impedance matching, Applications: In-Band Full Duplex (IBFD), Military, MIMO systems [6]

By carefully choosing the dielectric constant, both modes can be excited and decoupled simultaneously. The design was first validated on a two-element array and later extended to a four-element array. Moreover, the array supports beam scanning up to $\pm 46^\circ$ in the E-plane, which makes it useful for MIMO systems in modern wireless communication. Key Technical Points: Frequency band: 4.7–5.3 GHz, Antenna type: Dielectric Patch Antenna (DPA), Feeding technique: Microstrip-coupled slot feed on a Rogers substrate, Performance: 10.5% bandwidth, >21 dB isolation, scanning up to $\pm 46^\circ$. [7]

The design cleverly uses F-shaped slots with a slotted ground plane to achieve two independent resonances — one in the sub-6 GHz band (4.7–4.9 GHz, n79) and another in the mmWave band (28–29 GHz, n257). Unlike many earlier works that require external decoupling structures (like metasurfaces or parasitic elements), this design achieves high isolation just by the orthogonal placement of elements. It's neat because they managed to keep the antenna compact while still handling both low-band and high-band requirements for future smartphones. The results showed isolation better than 14.4 dB at sub-6 GHz and >25 dB at mmWave, with peak gains of 4.9 dBi (n79) and 7.2 dBi (n257). The antenna efficiency was also impressive, around 90%, which is quite high considering the compact size and the fact that there are 8 elements squeezed into a phone-sized board. Key Technical Points: Frequency bands: Sub-6 GHz (4.7–4.9 GHz, n79) and mmWave (28–29 GHz, n257), Antenna type: 8-element dual-polarised MIMO antenna (slot-based), Feeding technique: Single-fed with F-shaped feeding strips on a RO4003C substrate (0.8 mm thick), Performance: High isolation (>14.4 dB / >25 dB), peak gain (4.9 dBi / 7.2 dBi), and ~90% efficiency. [8]

A circularly polarised (CP) MIMO microstrip patch antenna (MPA) array that achieves self-decoupling without any extra decoupling structures. The authors made use of the mode counteraction effect. By exciting two orthogonal modes (TM₁₀ and TM₀₁) within the rectangular patch, the antenna not only generates circular polarisation but also creates a null-field region in the adjacent patch. When the feed of the second patch is placed in this null-field region, it cannot pick up unwanted coupled signals, resulting in an extremely low mutual coupling of about –59.4 dB at 5.5 GHz. The design uses a diagonal probe feed to excite both modes properly. The fabricated prototype worked very close to the simulations,

with only a small frequency shift (due to PCB tolerance). The results showed a 3 dB axial ratio bandwidth of $\sim 2.15\%$ (5.52–5.64 GHz), gain of ~ 4.8 dBi, and excellent port isolation. The authors also extended this approach to corner-truncated CP patch arrays and even a 1×4 MIMO array, which proved that the method is versatile and not limited to two-element arrays. Key Technical Points: Frequency band: ~ 5.45 – 5.63 GHz (centred at 5.5 GHz), Antenna type: Circularly Polarised Microstrip Patch Array (MIMO), Feeding technique: Coaxial probe (diagonal-fed), Performance: Isolation as low as -59.4 dB, 3 dB AR bandwidth $\sim 2.15\%$, gain ~ 4.8 dBi.[9]

This paper introduces a self-decoupling method for MIMO antenna arrays, which eliminates the need for extra decoupling structures. The technique relies on the characteristic modes (CMs) of the ground plane that naturally create null-field regions. When one antenna excites this mode, and another is positioned within the null field, high isolation (up to 57 dB) can be achieved. The method is versatile and suitable for different antenna types and array configurations. The core idea is to use the specific characteristic mode (Mode 2), which shows two null-field regions. Steps followed: Analyse the CMs of the ground plane using Altair Feko, identify the mode that generates parallel null regions (Mode 2 at 3.9–4 GHz), and excite this mode with one antenna. Place the second antenna's feed point in the null region.[10]

A practical and well-structured solution to one of the major challenges in compact MIMO antenna design—mutual coupling between closely spaced patch elements. Although MIMO technology is widely used to improve data rates and channel capacity in modern wireless systems, placing antenna elements very close to each other often leads to strong electromagnetic interaction. This interaction negatively affects impedance matching, radiation characteristics, and overall system efficiency. To address this issue, the authors introduce a simple yet effective decoupling technique based on a phase cancellation concept. The core idea behind the proposed method is that the coupled signal between adjacent patches has approximately a 90° phase shift. Therefore, if an additional signal path with a 270° phase shift and similar magnitude is introduced, the two signals cancel each other, resulting in improved isolation. Importantly, this structure is implemented on the same layer as the patch antennas, ensuring that the antenna remains low-profile and structurally simple without requiring additional layers or complex modifications. In addition to isolation enhancement, the antenna also showed better impedance matching and improved radiation

pattern symmetry. The technique was further extended to a three-element array, demonstrating that the approach is scalable for larger MIMO configurations without increasing complexity or thickness. Compared to other decoupling methods such as metasurfaces and electromagnetic bandgap structures, this design stands out due to its compact size, low profile, ease of implementation, and effective performance improvement, making it a highly practical solution for closely spaced MIMO patch antenna systems.[11]

In Simultaneous transmitting and receiving (STAR) systems, maintaining high isolation is extremely important. If the isolation is poor, unwanted signal leakage can occur, which may lead to receiver saturation and overall degradation of system performance. The difficulty becomes even greater in dual-polarised arrays, since coupling must be reduced in both polarisation directions without affecting radiation patterns, efficiency, or increasing the antenna size. To address this problem, the authors propose a simple and practical parasitic decoupling structure made up of a microstrip L-section array and a microstrip U-section array. The L-section mainly suppresses mutual coupling in the co-vertical polarisation (CVP), while the U-section targets the co-horizontal polarisation (CHP). The idea behind the design is straightforward: introduce an additional coupling path that produces a signal with nearly equal magnitude but opposite phase compared to the original coupling signal. When the phase difference approaches 180° , the two signals cancel each other through vector superposition, thereby improving isolation. By carefully adjusting the size and placement of these parasitic elements, effective decoupling can be achieved in both polarisations. To verify the concept, a 1×2 dual-polarised patch antenna array operating at 4.3 GHz was designed and fabricated. The measured results show a substantial improvement in isolation, reaching 37.5 dB in CHP and 33.6 dB in CVP, which is more than 17–18 dB better than the reference antenna. At the same time, the radiation gain remains nearly unchanged, and the total efficiency increases from 84% to 91% because the parasitic structure does not introduce additional insertion loss. The envelope correlation coefficient (ECC) is also reduced to below 0.004, indicating excellent diversity performance. Compared to traditional decoupling approaches such as defected ground structures, decoupling networks, metasurfaces, or superstrate techniques, this method offers a good balance of high isolation, high efficiency, compact size, and low profile. Since the parasitic elements are placed on the same layer as the antenna array, the overall thickness is not

increased. This makes the technique both practical and scalable for real-world dual-polarised antenna arrays used in modern communication and radar systems.[12]

Many conventional isolation techniques rely on complex decoupling structures, defected ground planes, or additional layers, which can increase design complexity and antenna profile. In contrast, this method keeps the structure simple by adding multiple square parasitic patches near the main radiating elements, improving performance without modifying the ground plane. When parasitic elements are introduced, part of the energy that would normally couple to the neighbouring antenna is instead attracted to these passive patches. This effectively reduces direct mutual coupling between the MIMO elements. At the same time, the parasitic patches create additional resonances that merge with the main resonance of the driven patch, leading to a noticeable improvement in impedance bandwidth. In a narrower frequency range, isolation greater than 40 dB is achieved. The realised gain also improves because the parasitic elements effectively increase the radiating aperture. Furthermore, the envelope correlation coefficient (ECC) remains low, indicating good diversity performance. Similar trends are observed in the E-plane configuration, where symmetrical placement of parasitic patches further enhances isolation and maintains radiation pattern symmetry. Another important advantage of this technique is its scalability. The method can be extended to multi-element arrays, particularly in the E-plane configuration, without disturbing the ground structure or significantly increasing the antenna height. Compared to many existing isolation enhancement methods, this approach offers a good balance between simplicity, low profile, wide bandwidth, and effective isolation improvement. [13]

The proposed MIMO antenna system operates within the sub-6 GHz frequency range, which is widely used for present-day broadband wireless services. It is designed to support multiple communication bands in this spectrum, making it suitable for LTE and upcoming 5G applications. The antenna maintains a reflection coefficient ($|S_{11}|$) below -10 dB throughout the operating band, indicating proper impedance matching and efficient power transfer from the feed to free space. The wide bandwidth is not accidental; it is achieved through systematic optimization of the patch dimensions along with the inclusion of a parasitic decoupling element that helps stabilize performance across frequencies. One of the main design challenges addressed in this system is the close spacing between antenna elements. To keep the overall structure compact, the two radiating patches are placed much

closer than half a wavelength at the center frequency. Normally, such tight spacing would result in strong mutual coupling due to overlapping near fields. However, this issue is controlled by introducing a thin parasitic strip between the facing radiating edges. Positioned in the region of strongest electromagnetic interaction, this strip modifies the surface current distribution and reduces energy transfer from one element to the other. As a result, the isolation remains better than -20 dB over most of the operating band, even with minimal separation. Regarding radiation performance, the antenna array maintains stable and predictable radiation characteristics across the entire bandwidth. Each element primarily radiates in the broadside direction, and the overall pattern remains nearly omnidirectional in the azimuth plane, which is desirable for mobile and portable devices. Pattern distortion due to the presence of the adjacent element is minimal, demonstrating that the decoupling structure does not negatively affect radiation behaviour. The realised gain remains reasonably consistent across the band, and radiation efficiency remains high, indicating that the isolation technique does not introduce significant additional losses. From a MIMO system perspective, the antenna exhibits excellent diversity performance. The envelope correlation coefficient (ECC) is maintained at a very low value, typically below 0.05, which implies that the signals received or transmitted by the two ports are largely uncorrelated. This directly improves channel capacity and link reliability in multipath environments. The diversity gain is close to its theoretical maximum, further confirming the effectiveness of the design. Overall, the combination of wide bandwidth, compact element spacing, strong isolation, and stable radiation performance makes this antenna configuration well-suited for practical sub-6 GHz MIMO communication systems. [14]

The frequency band (5.850–5.925 GHz) is widely allocated worldwide for vehicular communication systems. Each radiating element is a square microstrip patch with truncated opposite corners, a simple yet effective technique used to generate right-hand circular polarization (RHCP). To realise MIMO operation, four identical CP elements are arranged orthogonally on a common substrate. This orthogonal placement provides polarization diversity, which improves signal robustness in multipath vehicular environments. The minimum spacing between elements is only 8.65 mm, making the design compact relative to the operating wavelength. However, placing antenna elements so close to each other naturally increases mutual coupling, which can reduce isolation and negatively affect system capacity. In contrast, the circular ring parasitic structures positioned between the

radiators are responsible for suppressing coupling between perpendicularly placed elements. When these two techniques are combined, a substantial improvement in isolation is achieved. The isolation level increases from about 21–28 dB (without any decoupling) to more than 34 dB, reaching values as high as 65 dB for adjacent elements. Surface current distribution analysis clearly shows that the decoupling structures effectively block unwanted current propagation toward neighboring patches, confirming the reduction in electromagnetic interaction. Importantly, the impedance bandwidth and circular polarization performance remain stable even after introducing the decoupling structures. The diversity performance is excellent, with an envelope correlation coefficient (ECC) below 0.001 and a diversity gain close to the ideal 10 dB. The channel capacity loss is maintained below 0.15 bits/Hz, indicating minimal performance degradation in MIMO operation. Additionally, link budget analysis demonstrates that the antenna can support reliable V2X communication over distances of up to 1.5 km with a transmit power of 21 dBm, confirming its suitability for real-world vehicular communication applications.[15]

A compact two-element microstrip antenna array designed to operate at 2.4 GHz, with the primary goal of reducing mutual coupling between closely spaced elements. In practical wireless systems, especially in compact devices, antenna elements are often positioned very close to each other due to space limitations. This proximity leads to strong electromagnetic interaction caused by surface waves and near-field coupling, which can negatively affect impedance matching, radiation efficiency, and overall antenna performance. To overcome this issue, the authors introduce a straightforward and effective solution by incorporating two parasitic microstrips above the radiating patches. The antenna array consists of two rectangular microstrip patches fed using 50-ohm coaxial probes. These patches are printed on a dielectric substrate with a thickness of 3.2 mm and a relative permittivity of 2.2. Above the patches, an additional thin dielectric layer is placed, on which two parasitic microstrips are etched. These strips are not directly connected to the feed; instead, they function through electromagnetic coupling with the radiating elements. By carefully optimizing key structural parameters such as the strip length, the spacing between patches, and the thickness of the upper dielectric layer, the design achieves effective suppression of mutual coupling while maintaining resonance at the desired 2.4 GHz frequency. When the two parasitic strips are introduced, the resonant frequency shifts slightly to 2.4 GHz, and the mutual coupling improves significantly to approximately -31.2 dB. A detailed parametric study further

shows that the thickness of the upper layer (h_2), the strip length (a_1), and the spacing between the elements (d) strongly influence both the coupling level and the resonant behavior. The measured results closely follow the simulated predictions, confirming the reliability of the design. At 2.4 GHz, the measured mutual coupling is below -32 dB, and the reflection coefficient at resonance is better than -30 dB. Radiation pattern measurements in both the x - z and y - z planes indicate that the parasitic microstrips have minimal impact on the overall radiation characteristics. The maximum measured gain remains around 6.2–6.4 dB, showing that the technique effectively reduces mutual coupling without compromising antenna gain or radiation performance.[16]

When the reflection responses of the CM and DM modes are made nearly identical at a specific centre frequency, the coupling components cancel each other, resulting in a significant reduction in mutual coupling between the ports. To realize this condition, a decoupling probe is introduced along the symmetry plane of the antenna structure. This probe effectively modifies the common-mode impedance while leaving the differential-mode characteristics largely unaffected. By carefully optimizing the probe position and dimensions, strong isolation is obtained near 6.03 GHz, where the antenna achieves a peak isolation level of approximately 42 dB. In this initial configuration, the antenna operates with an impedance bandwidth of roughly 5.71 GHz to 6.31 GHz, demonstrating effective decoupling around the target frequency. After achieving high isolation at the center frequency, independent matching networks are introduced to shift the operating frequencies of the two ports to adjacent communication bands. These matching circuits are implemented using microstrip transmission lines combined with lumped inductors of approximately 1.5 nH and 0.7 nH, allowing each antenna port to be tuned independently without significantly affecting the established decoupling condition. With this configuration, the simulated -10 dB impedance bandwidth for Port 1 extends from approximately 5.41 GHz to 6.07 GHz, while Port 2 operates over a wider frequency range from around 5.58 GHz to 7.49 GHz. To further improve the isolation across the entire operating bandwidth, bandpass filters are integrated into the feeding network. These filters suppress unwanted signal leakage near the band edges and help maintain consistent isolation between the ports across the full operating frequency range. The measured gain shows slight variations across the frequency range due to the insertion losses introduced by the filters, which are approximately 1.7 dB and 2.3 dB, respectively. Nevertheless, the antenna maintains satisfactory radiation efficiency and

consistent directional radiation characteristics, demonstrating its suitability for compact multi-band wireless communication systems .[17]

From the radiation perspective, the antenna exhibits stable radiation patterns with noticeable pattern diversity between the two ports. Owing to the symmetrical configuration of the radiating structure and the feeding network, both ports present nearly similar radiation characteristics across the operating frequency band. The measured antenna gain varies from approximately 2 dBi to 3.14 dBi, with the maximum measured gain reaching about 3.14 dBi, while the simulated peak gain is around 2.9 dBi. In addition, the radiation patterns show slight variations in the main beam direction for the two ports, which helps improve spatial diversity and enhances signal reliability in multi-antenna wireless communication environments. The antenna also maintains good impedance matching and satisfactory radiation efficiency across the operating band while preserving a compact physical structure. The integration of shorting pins and metallic strips for decoupling effectively suppresses the coupling currents without requiring complex ground plane modifications or bulky isolation structures. This approach helps maintain a simple and compact antenna geometry while still achieving high isolation performance. Overall, the antenna demonstrates a balanced combination of high isolation (greater than 25 dB), moderate gain of around 3 dBi, stable radiation characteristics, and efficient impedance matching, making it well suited for compact sub-6 GHz wireless and MIMO communication devices where mutual coupling reduction and consistent radiation performance are critical design requirements.[18]

3.Methodology

3.1 Overview

The design process began with a single-element microstrip patch antenna using ANSYS HFSS simulation software. The antenna geometry, substrate type, and feeding mechanism were carefully selected to achieve resonance at the desired frequency band. Once the basic element was designed, its performance was analysed in terms of S-parameters and gain plots to ensure good impedance matching and radiation efficiency. Based on this single-element analysis, a 1×2 MIMO antenna array was designed without any decoupling structure to study the mutual coupling behaviour between the antenna elements and understand its impact on system performance. After identifying coupling issues, a suitable decoupling structure was inserted into the MIMO configuration to enhance isolation and improve overall gain performance. The optimised antenna design was then fabricated using appropriate PCB materials and standard microstrip fabrication techniques. The fabricated prototype underwent experimental validation using a vector network analyser and anechoic chamber measurements to compare simulated and measured results. Finally, all observations, simulation data, and test outcomes were compiled and documented to evaluate the antenna's effectiveness and verify the design objectives.

3.2 Software Setup

The design and simulation work for the antenna were carried out using ANSYS HFSS (High-Frequency Structure Simulator), which is a 3D electromagnetic simulation tool widely used for antenna modelling and analysis. The setup process began by defining the operating frequency at 2.4 GHz, which lies within the ISM (Industrial, Scientific, and Medical) band commonly used for Wi-Fi and Bluetooth applications. The simulation environment was configured with appropriate solution frequency and frequency sweep range to ensure accurate results around this band.

A suitable substrate material (such as Rogers RT/Duroid) was selected within HFSS, and its dielectric constant, loss tangent, and thickness were specified. The ground plane, patch, and feedline were modelled according to the calculated dimensions for 2.4 GHz resonance.

Radiation boundaries and wave ports were assigned to simulate realistic free-space conditions and signal excitation. The model was then solved using the HFSS setup wizard, and key parameters such as S_{11} (return loss), gain, and radiation pattern were analysed. This setup provided a reliable simulation environment for both single-element and MIMO antenna configurations, helping to predict and optimise performance before fabrication.

Simulation setup:

Solution frequency: 2.4GHz, Substrate: Rogers RT duroid 5880 ($\epsilon_r = 2.2$), Substrate height: 1.6mm.

3. Antenna Design

The antenna is designed such that it operates at 2.4 GHz, a frequency Wi-Fi, Bluetooth, and ISM band applications. A rectangular microstrip patch antenna configuration was chosen because of its compact size, ease of integration with circuits, and reliable directional radiation characteristics. For better performance, the design utilised Rogers RT/Duroid 5880 as the substrate material due to its low dielectric constant ($\epsilon_r = 2.2$) and low loss tangent ($\tan \delta = 0.0009$).

The return loss of the patch antenna is determined by the width W. The frequency of operation of the patch antenna is determined by the length L. The centre frequency is given by:

$$f = \frac{c}{2L\sqrt{\epsilon_r}} = \frac{1}{2L\sqrt{\epsilon_0\epsilon_r\mu_0}} \quad (4.1)$$

Width of the patch is given by :

$$W_p = \frac{c}{2f_0\sqrt{\frac{\epsilon_r+1}{2}}} \quad (4.2)$$

$$W = \frac{3 \times 10^8}{2(2.4 \times 10^9)} \sqrt{\frac{2}{2.2+1}} = 49.38 \text{ mm} \quad (4.3)$$

Figure 4.1- Simulated model of a unit element of an Array

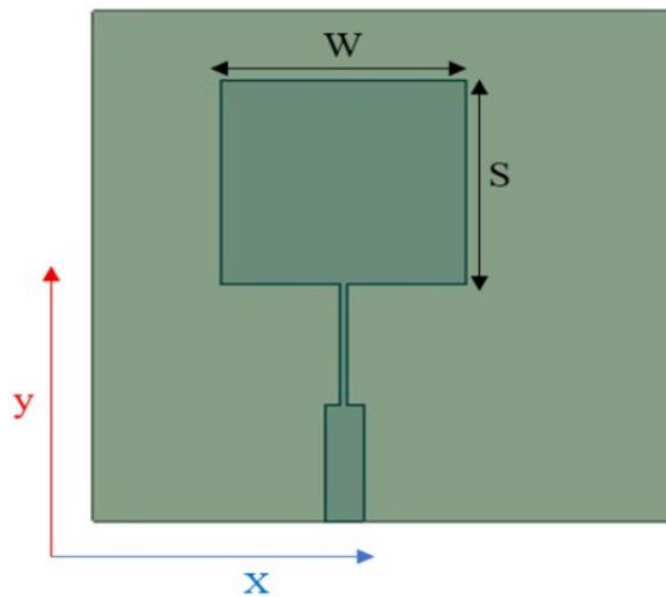
Effective dielectric constant (ϵ_{eff}):

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \sqrt{\frac{1}{1 + 12 \frac{h}{W_p}}} \quad (4.4)$$

$$\frac{W}{h} = \frac{49.38}{1.6} = 30.86 \quad (4.5)$$

$$\epsilon_{eff} = 2.03 \quad (4.6)$$

The



extension length:

$$\Delta L = 0.412h \frac{(\epsilon_{eff}+0.3)(\frac{W_p}{h}+0.264)}{(\epsilon_{eff}-0.258)(\frac{W_p}{h}+0.8)} \quad (4.7)$$

$$\Delta L = 0.74 \text{ mm} \quad (4.8)$$

Effective length (L_{eff}):

$$L_{eff} = \frac{c}{2f_0\sqrt{\epsilon_{eff}}} \quad (4.9)$$

$$L_{eff} = 30.2 \text{ mm} \quad (4.10)$$

Physical length of the patch is given by:

$$L = L_{eff} - 2\Delta L \quad (4.11)$$

$$L = 28.7 \text{ mm} \quad (4.12)$$

Impedance matching for quarter-wave transformer feeding:

$$Z_t = \sqrt{Z_0 \times Z_{in}} \quad (4.13)$$

$$Z_{in} = \frac{40(\epsilon_r)^2}{\epsilon_r - 1} \left(\frac{L}{W}\right)^2 \quad (4.14)$$

$$Z_0 = 50 \Omega, Z_{in} = 100 \Omega, Z_t = 70.7 \Omega$$

$$L_t = \frac{\lambda_g}{4} \quad (4.15)$$

$$\lambda_g = \frac{c}{f_r\sqrt{\epsilon_{eff}}} \quad (4.16)$$

$$L_t = 30.8 \text{ mm}$$

Here, the length is $\lambda_0/4$, and the width can be calculated as follows:

If $\left(\frac{W}{H}\right) > 1$:

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}} \left(\frac{W}{H} + 1.393 + \frac{2}{3} \ln\left(\frac{W}{H} + 1.444\right) \right)} \quad (4.17)$$

Using Equations 4.4 and 4.17 Width can be calculated.

If $\left(\frac{W}{H}\right) < 1$:

$$Z_0 = \frac{60}{\sqrt{\epsilon_{eff}}} \ln \left(8 \left(\frac{H}{W} \right) + 0.25 \left(\frac{W}{H} \right) \right) \quad (4.18)$$

Using Equations 4.4 and 4.18 Width can be calculated.

Table 4-1-Calculated Antenna Dimensions for single element

Variables	Value
W	49.38 mm
L	28.7 mm
W_g	70 mm
L_g	70 mm
h	1.6 mm
Z_0	50 Ω
Z_t	70.7 Ω
L_t	30.8 mm
W_f	4.9 mm
W_t	1.09 mm

The patch width (W) is calculated with the help of operating frequency and dielectric constant to control the resonant behaviour and radiation efficiency, resulting in a width of 49.8 mm. To account for the effect of fringing fields at the edges, an effective dielectric constant of 2.03 is used to compute the effective length of the patch. The length extension (ΔL), caused by fringing, is calculated to be 0.74 mm, which leads to an actual patch length (L) of 28.7 mm.

A 50 Ω microstrip line is used to match standard transmission lines. To efficiently match the patch input resistance with the feedline, a quarter-wave transformer is inserted. Its characteristic impedance (Z_t) is calculated as 70.7 Ω , and its length (L_t) is a quarter of the guided wavelength (≈ 30.8 mm) based on the effective dielectric constant. The

corresponding microstrip widths are computed using standard transmission line equations, giving $W_f = 4.9$ mm for the $50\ \Omega$ line and $W_t = 1.09$ mm for the $70.7\ \Omega$ transformer line

4.1 Two-element Array

To upgrade the performance of the single patch antenna, particularly its gain and directivity, a two-element linear array is designed using the same rectangular microstrip patch on Rogers RT/duroid 5880. Essentially, two identical patches, each simulated to resonate at 2.4 GHz, and arranged side by side along a straight line. The goal of this setup is to have the two elements work together to create a stronger, more focused radiation in the broadside direction while keeping the impedance properly matched. Each patch is fed using its own quarter-wave transformer, which connects to a $50\ \Omega$ feedline. This ensures that the patches receive the correct amount of power and remain well-matched to the source. The spacing between the centres of the two patches is chosen to be approximately $0.5\ \lambda_0$ at 2.4 GHz. This spacing is ideal because it helps the radiated waves from the two patches add constructively in the main lobe direction, enhancing the overall gain without creating unwanted grating lobes, also increasing isolation between the patches.

The patch dimensions, transformer parameters, and substrate properties remain exactly the same as the single patch design. The $50\ \Omega$ feedline splits into two paths, each feeding one patch through its quarter-wave transformer. The transformers maintain proper impedance matching despite the two patches being connected, so the input sees a clean $50\ \Omega$ load.

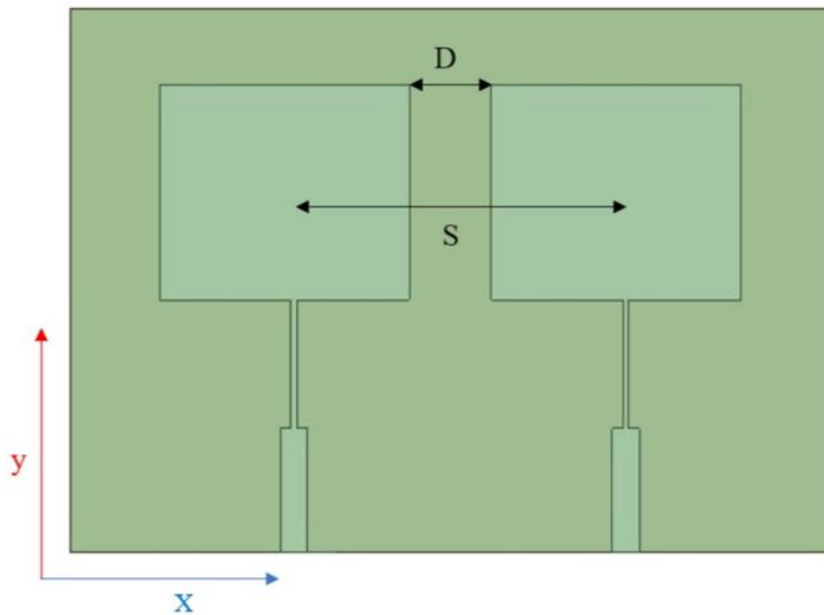


Figure 4.2-Two-element array

Table 4-2 – Spacing between the antennas

Variable	Value	Value in terms of λ_0
D(edge to edge)	24.6 mm	$0.2 \lambda_0$
S(centre to centre)	71.4mm	$0.56 \lambda_0$

4.2 Design of a two-element patch antenna array incorporating a neutralisation line

4.2.1 Neutralisation Line (NL)

A neutralisation line (NL) is a thin metallic strip (microstrip conductor) placed between two antenna elements, usually connecting their feed lines or positioned near the radiating edges. Its purpose is to intentionally create an additional coupling path between the antennas.

The neutralisation line is designed in such a way that it carries a current equal in magnitude but opposite in phase to the unwanted coupled current. Because the two currents are 180° out of phase, they cancel each other through destructive interference. As a result, the overall coupled energy reaching the adjacent antenna port is significantly reduced.

When one antenna is excited, electromagnetic fields and surface currents induce a coupling current in the neighbouring antenna. The neutralisation line provides an alternative path for this interaction. Due to its carefully chosen electrical length, the current flowing through the NL has an opposite phase compared to the naturally induced coupling current.

When both currents combine at the receiving port, they cancel each other, thereby improving isolation (reducing S21).

4.2.2 Neutralisation Line Length Calculation

4.2.2.1 For 180° phase shift:

$$L_{NL} \approx \frac{\lambda_g}{2}$$

4.2.2.2 Where guided wavelength:

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}}$$

4.2.2.3 Free-space wavelength:

$$\lambda_0 = \frac{c}{f_0} = 125 \text{ mm}$$

$$\lambda_g = \frac{125}{\sqrt{2.08}}$$

$$\lambda_g \approx 86.7 \text{ mm}$$

4.2.2.4 Neutralisation Line Length:

$$L_{NL} \approx \frac{86.7}{2}$$

$$L_{NL} \approx 43 \text{ mm}$$

4.2.2.5 Neutralisation Line Width:

The width controls coupling strength.

Start with:

$$W_{NL} = 0.5 \text{ mm} - 1 \text{ mm}$$

4.2.2.6 Placement Guidelines

- Connect between the two feed lines (near the inset feed region)
- Or connect between patch edges where the surface current is maximum
- Maintain symmetry

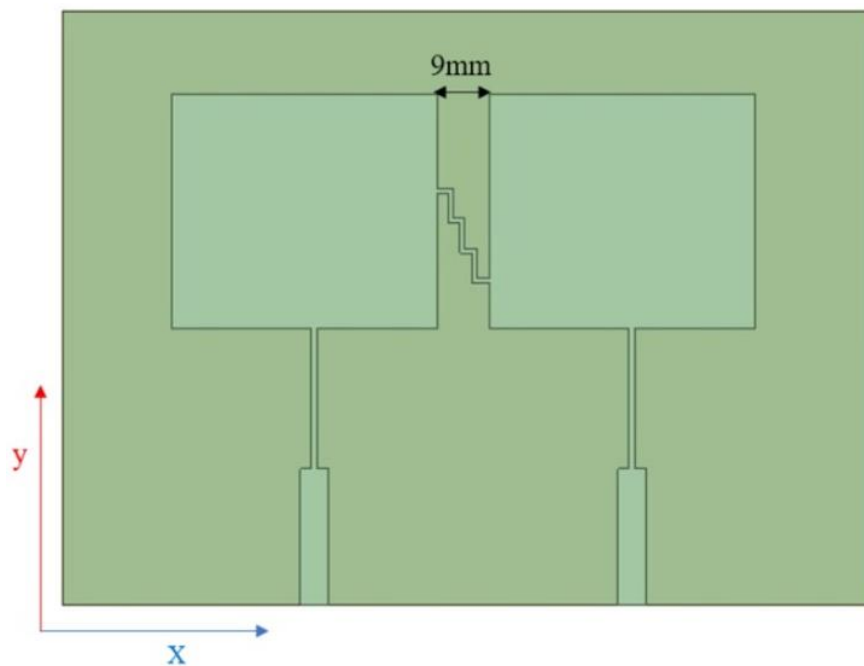


Figure 4.3-Two-element array with neutralisation line

4.3 Design of a two-element patch antenna array incorporating a parasitic element

4.3.1 Parasitic Elements

The parasitic element is designed with a specific length so that it creates a controlled electrical path between the two antenna elements. When one patch is excited, coupling currents try to flow toward the adjacent patch. The parasitic element intercepts this energy and becomes excited by the nearby electromagnetic field.

Because its length is carefully chosen (close to resonance at the operating frequency), the current induced on the parasitic element develops with a phase that is opposite to the unwanted coupling current between the patches.

As a result, the two currents interfere destructively. This cancellation reduces the net coupled energy reaching the second antenna, thereby improving isolation at the design frequency.

4.3.2 Parameters:

Free space wavelength: $\lambda_0 = 125\text{mm}$

Length of the parasitic element = $47.5\text{ mm} = 0.38\lambda_0$

Width = 1.5mm

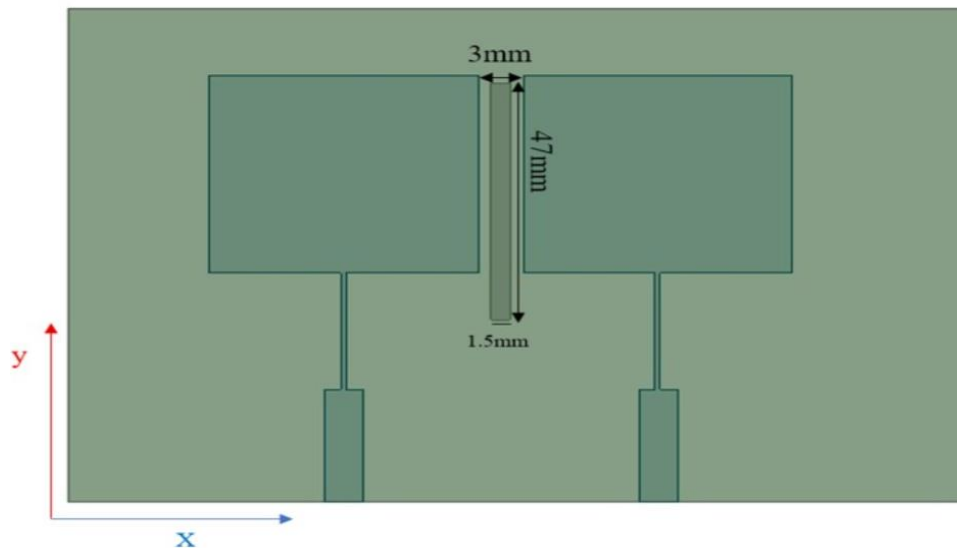


Figure 4.4-Two-element array with parasitic element

4.3.3 Final proposed structure- Parasitic element with anti-parallel antenna configuration

Between the two rectangular patches, a narrow parasitic strip is placed vertically in the small gap that separates them. The patches are positioned face-to-face, so their radiating edges directly confront each other. Because of this close spacing, the electromagnetic fields at the edges do not remain confined to a single patch. Instead, they spread into the gap region and naturally interact with the opposing patch. This creates a strong coupling path between the two elements. The parasitic strip sits right in this interaction zone, where the electric field intensity is relatively high. Although it is not directly excited, it responds to the surrounding fields and supports induced currents. As a result, it alters how energy is exchanged between the two patches and slightly reshapes the current distribution along their edges.

From a structural perspective, this face-to-face arrangement makes the system compact while intentionally allowing interaction between the radiators. The parasitic strip acts as a passive tuning element within this shared field region. By adjusting its length, width, or spacing from the patches, the coupling level can be controlled, and small shifts in resonant frequency or bandwidth can be achieved. In simple terms, the two patches form the primary radiating elements, and the central strip fine-tunes how strongly they influence each other. The overall design relies on this controlled interaction to achieve the desired electromagnetic performance.

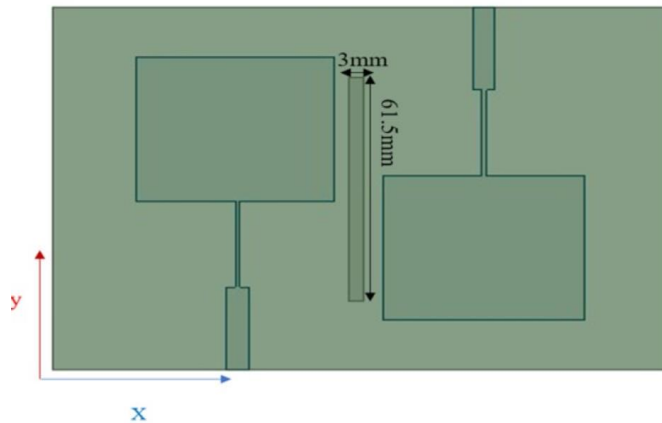


Figure 4.5- Final proposed structure-Parasitic element with anti-parallel antenna configuration

5. Results and Discussion

5.1 Performance analysis of a single-element antenna

The designed rectangular microstrip patch antenna was analysed to assess its performance at the target frequency of 2.4 GHz, which is part of the ISM band. From a theoretical perspective, a microstrip patch resonates when the length of the patch corresponds to approximately half a wavelength in the dielectric medium.

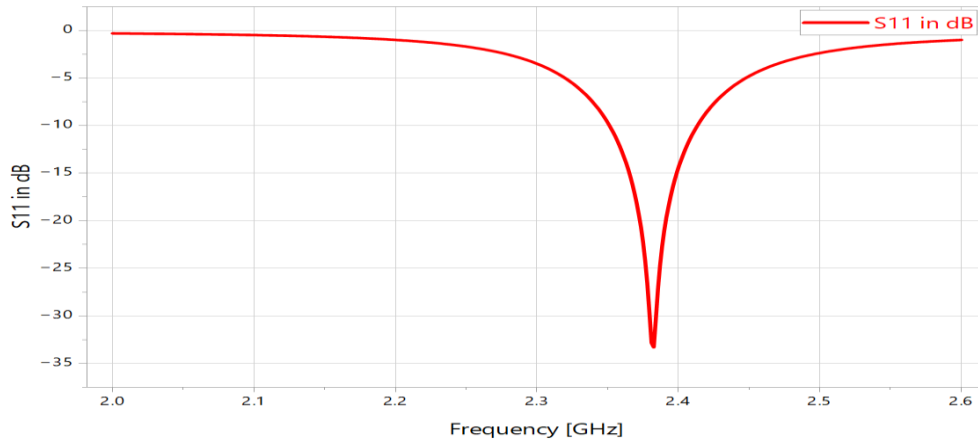


Figure 5.1-Simulated S-parameter of a single element

As shown in Figure 5-1, antenna design has sufficiently good input matching (where $|S_{11}| = -10$ dB) at 2.4 GHz (-27.33 dB).

5.2 Performance analysis of Two-element array

The two-element linear array was designed by extending the single rectangular patch antenna described earlier. From a theoretical perspective, a linear array combines the fields of multiple identical elements to enhance gain and directivity. In our case, the two patches are placed side by side along a straight line, with a centre-to-centre spacing of more than $0.5 \lambda_0$ (≈ 62.5 mm at 2.4 GHz). This spacing is chosen based on array theory to ensure constructive interference in the broadside direction while avoiding significant grating lobes.

Both patches are fed through individual quarter-wave transformers, which are connected to a common 50Ω feedline. The transformers ensure proper impedance matching for each element, theoretically transforming the patch input impedance to match the feedline and minimise reflection losses. The simulated S_{11} shows a strong resonance at 2.4 GHz, with values below -20 dB, confirming that the array is well-matched at the target frequency.

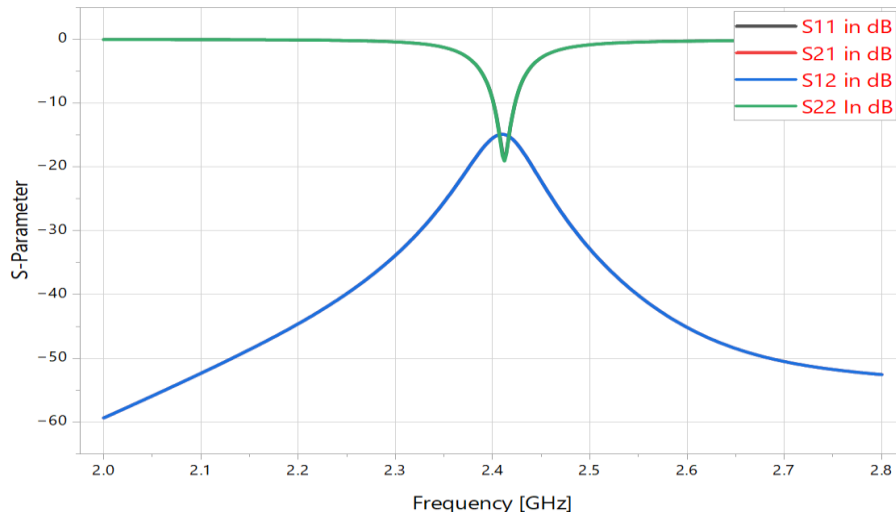


Figure 5.2-Simulated S-parameter plot of a two-element array

5.3 Performance analysis of Two-element array with a parasitic element

In the basic configuration, the two patch antennas are placed very close to each other with a spacing of 7mm ($0.056\lambda_0$). Without any parasitic element, the antenna resonates around 2.389 GHz with a return loss of about -16 dB. Even though the operating frequency is near the required 2.4 GHz band, the mutual coupling is quite high (around -7.5 dB), which shows that the two patches are strongly interacting because of the small separation.

After inserting the parasitic strip between the two antennas, the behaviour of the antenna changes noticeably. The resonant frequency shifts closer to 2.4 GHz (around 2.404 GHz), and the return loss improves to approximately -24.6 dB, indicating better impedance matching.

At the same time, the mutual coupling reduces to nearly -13.2 dB. This reduction clearly shows that the parasitic strip is helping to control the unwanted interaction between the two elements. It modifies the surface current distribution and weakens the coupling path between the patches, even though the spacing remains only 5 mm.

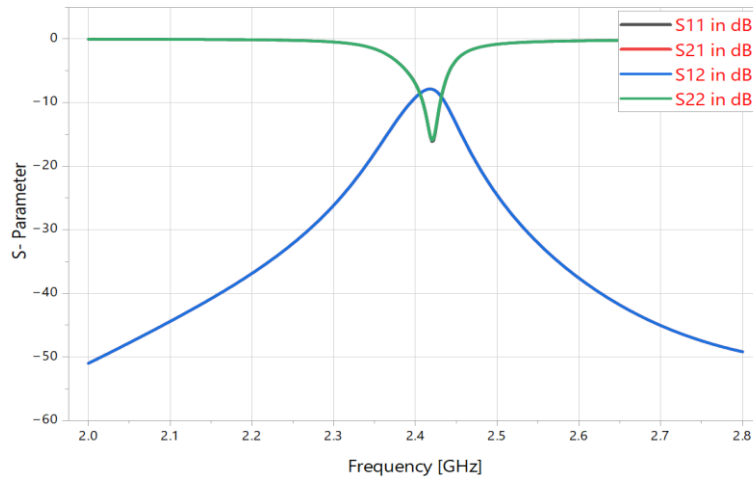


Figure 5.3.a- Simulated S parameter of a 1×2 Patch antenna Array without a parasitic element

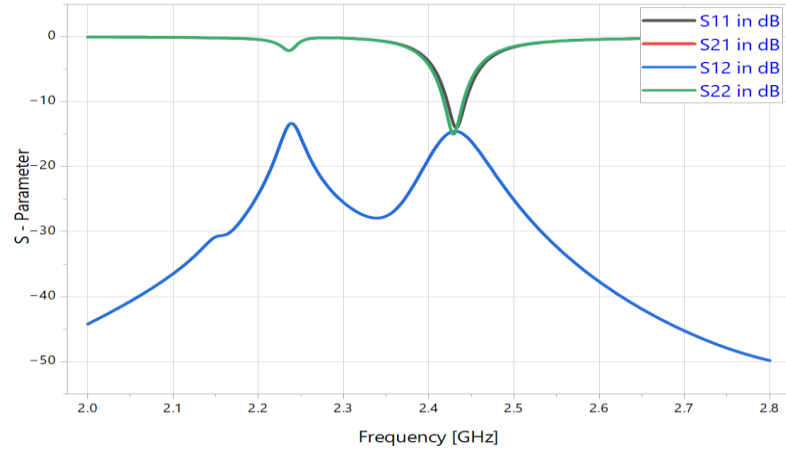


Figure 5.3.b- Simulated S parameter of a 1×2 Patch antenna Array with a parasitic element

5.3.1 Performance analysis of the final proposed structure-parasitic element with anti-parallel antenna configuration

After inserting and carefully optimising the parasitic strip between the two patches, the final S-parameter response (as seen in the attached plot) clearly shows a noticeable improvement in performance.

From the final graph, the antenna now resonates at approximately 2.428 GHz, which lies well within the required 2.4 GHz band. The return loss ($S_{11} \approx -22.7$ dB) indicates strong impedance matching, meaning the antenna is efficiently radiating with minimal reflected power.

More importantly, the mutual coupling ($S_{21} \approx -17.38$ dB) is significantly reduced. This is a major improvement compared to the earlier stages of the design:

- Initial coupling: -7.5 dB (very high interaction)
- After initial modification: ~ -13 dB
- Final optimised structure: -17.4 dB

This represents nearly a 10 dB improvement in isolation compared to the original configuration.

In simple terms, an S_{21} value of 17.4 dB means that only a small portion of the power from Port 1 is reaching Port 2. Most of the energy remains confined to the excited antenna, which confirms that the parasitic strip is effectively reducing the unwanted interaction between the two patches even though the spacing is still only 7 mm.

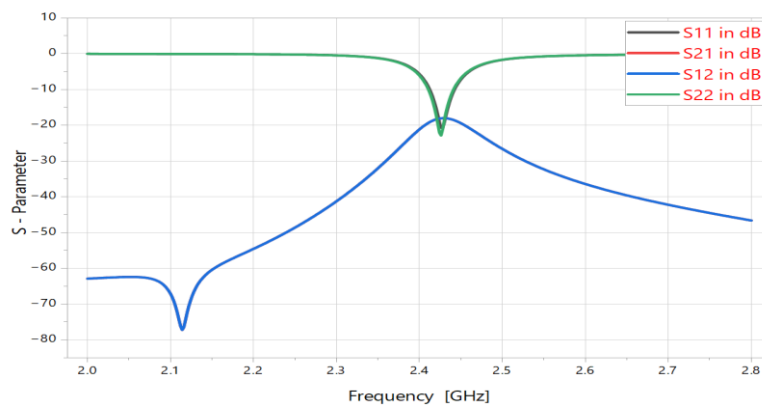


Figure 5.3.1.a- Simulated S parameter of a 1×2 Patch antenna Array without a parasitic element with anti-parallel antenna configuration

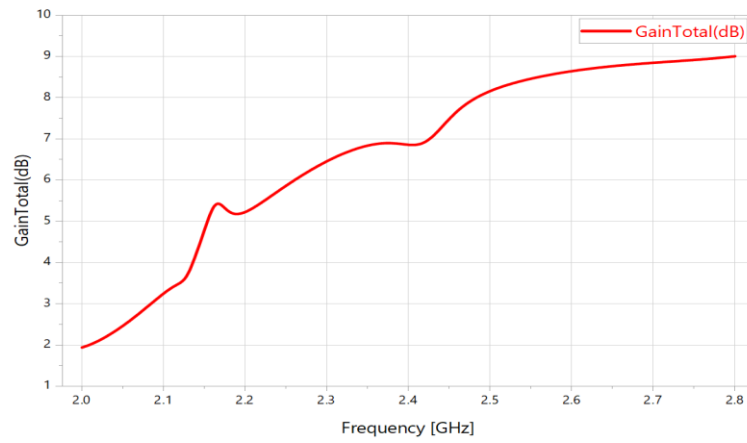


Figure 5.3.1.b- Simulated Gain plot of a 1×2 Patch antenna Array with a parasitic element with anti-parallel antenna configuration

5.4 Comparison of results

In the initial configuration, the two patch antennas were placed very close to each other with a spacing of about 7 mm ($0.056\lambda_0$) and no decoupling structure was used. Under this condition, the mutual coupling between the antenna elements was observed to be around -7.5 dB. This relatively high coupling value indicates a strong electromagnetic interaction between the two antennas due to their very small spacing.

To reduce this coupling effect, a neutralisation line was introduced between the two antenna elements. With this modification, the simulated S_{21} value improved to approximately -11.69 dB at 2.44 GHz. Although this shows a reduction in coupling compared to the initial design, the isolation level is still lower than the typical requirement of 20 dB for high-performance MIMO antenna systems.

Further improvement was achieved by introducing a parasitic element between the two patches. The parasitic strip alters the surface current distribution and helps weaken the coupling path between the antenna elements. As a result, the S_{21} value reduced to around -13.2 dB, showing a noticeable improvement in isolation.

Finally, after optimising the parasitic element structure, the antenna achieved a significantly better isolation of approximately -17.38 dB at 2.428 GHz. This corresponds to nearly a 10 dB improvement compared to the initial configuration, demonstrating that the optimised parasitic structure effectively suppresses the coupling currents between the antenna elements while maintaining a compact antenna arrangement.

Table 5-1 Comparison of results

Method of mutual coupling reduction	S_{21} without modifying the two-element array structure(in dB)	S_{21} after modifying the two-element array structure (in dB)
Neutralisation line	7.5	11.7
Parasitic element	7.5	13.2
Final proposed structure: Parasitic element with anti-parallel antenna configuration	7.5	17.4

6. Conclusion

In this project, a compact 1×2 microstrip patch antenna array operating in the 2.4 GHz ISM band was designed for Wi-Fi and Bluetooth applications. The primary challenge addressed was the mutual coupling that occurs when antenna elements are placed close to each other in a compact array configuration. To mitigate this effect, a parasitic decoupling structure was introduced between the two patch elements.

The antenna structure was designed and optimised using Ansys HFSS, where parameters such as the geometry of the parasitic element and the spacing between the patches were carefully tuned. Simulation results demonstrated good impedance matching near 2.4 GHz, along with a significant reduction in mutual coupling. The final design achieved an isolation of approximately 17.4 dB between the antenna elements while maintaining stable radiation characteristics. In addition, the antenna array produced a broadside radiation pattern with a gain of around 7 dBi, making it suitable for reliable short-range wireless communication.

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